

Linux commands

- ls
 - ls - list the contents of present working directory
 - ls path - list the contents of given path
 - ls -l - list the contents in detail format
 - type and permissions
 - Types of files
 - Permissions of files
 - r - read, w - write, x - execute
 - (rwx)user/owner, (rwx)group, (rwx)others
 - link count
 - user/owner
 - group
 - size
 - timestamp
 - name
 - ls -a - list all contents along with hidden
 - ls -A - list all contents along with hidden except . and ..
 - ls -i - list contents with inode number
 - inode number is unique number given to every file
 - ls -s - list content with size (number of blocks)
 - ls -S - list content in descending order of their sizes
- stat
 - stat file - display information of file
 - stat file1 file2 - display information of file1 and file2
 - stat -c "format" file - display file information in given format
- head
 - head file - display first 10 lines
 - head -5 file - display first 5 lines
- tail
 - tail file - display last 10 lines
 - tail -4 file - display last 4 lines
- sort
 - sort file - sort the content by alphabetically
 - sort -n file - sort the content by their value
 - sort command do not modify file content
- uniq
 - uniq file - display contents uniquely (truncate duplicate)

- truncate duplicate content if it is consecutive
- rev filepath
 - Print each line reversed.
 - File contents are not modified.
- tac filepath
 - Print all lines in reverse order. The first line printed at last, while last line printed first.
- alias
 - alias list="ls -l"
 - list will be alias/nick name to ls -l
 - list will give output same as ls -l
- unalias
 - unalias list
- which
 - which command
 - display the location of command executable.
- whereis
 - whereis command
 - display the location of command executable and also manual page location.

Redirection

- for every command input is taken from terminal, output is printed on terminal and error is also printed on terminal
- Standard streams (by default for every process, three files are opened)
 - stdin
 - stdout
 - stderr
- There are three types of redirections
 - input redirection
 - input will be taken from file instead of stdin
 - to do input redirection '<' symbol is used
 - command < file
 - output redirection
 - output will be written into file instead of stdout
 - to do output redirection '>' or '>>' symbol is used
 - command > file
 - older content of file will be over written

- `command >> file`
 - content will be appneded into file at the end
- error redirection
 - error will be written into file instead of stderr
 - to do output redirection '2>' or '2>>' symbol is used
 - `command 2> file`
 - older content of file will be over written
 - `command 2>> file`
 - content will be appneded into file at the end

Pipe

- Using pipe, we can redirect output of any command to the input of any other command.
- Two processes are connected using pipe operator (`|`).
- Two processes runs simultaneously and are automatically rescheduled as data flows between them.
- If you don't use pipes, you must use several steps to do single task.
- `command1 | command2`
 - output of command1 will be given as input to command 2
- E.g.
 - `who | wc`

Shell meta characters

- '*' - zero or more occurances of any character
- '?' - one occurrence of any character

Regular Expressions

- Find a pattern in text file(s).
- Regular expressions are patterns used to match character combinations in strings.
- A regular expression pattern is composed of simple characters, or a combination of simple and special characters e.g. `/abc/`, `/ab*c/`
- Pattern is given using regex wild-card characters.
 - Basic wild-card characters
 - `$` - find at the end of line.
 - `^` - find at the start of line.
 - `[]` - any single char in give range or set of chars
 - `[^ ]` - any single char not in give range or set of chars
 - `.` - any single character
 - `*` - zero or more occurrences of previous character
 - Extended wild-card characters
 - `?` - zero or one occurrence of previous character
 - `+` - one or more occurrences of previous character
 - `{n}` - n occurrences of previous character
 - `{,n}` - max n occurrences of previous character
 - `{m,}` - min m occurrences of previous character

- {m,n} - min m and max n occurrences of previous character
- () - grouping (chars)
- (|) - find one of the group of characters

grep

- Regex commands
 - grep - GNU Regular Expression Parser - Basic wild-card
 - egrep - Extended Grep - Basic + Extended wild-card
 - fgrep - Fixed Grep - No wild-card
- Command syntax
 - grep "pattern" filepath
 - grep [options] "pattern" filepath
 - -c : count number of occurrences
 - -v : invert the find output
 - -i : case insensitive search
 - -w : search whole words only
 - -R : search recursively in a directory
 - -n : show line number.

File Permissions/Mode

- File permission types: r (read), w (write), x (execute)
- Permission levels: u (user), g (group), o (other)
- File mode: u-rwx g-rwx o-rwx
- Example: hello.txt -- user can read/write and execute, group can read and execute, other can only read.
 - hello.txt mode: u=rwx g=r-x o=r--
 - rwx r-x r--
 - 111 101 100 = 754